WAC 3.0: from Windows XP to Windows 11, and what the disk kept

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The article on version 2.0 presented WAC (Windows Artefacts Collector): a free collector, in C++, that reads the volumes raw, converts to JSON on the analysis workstation, and seals and logs every one of its operations, writing nothing on the examined machine. Version 3.0 keeps these principles and widens their reach: every Windows workstation since XP SP1, shadow copies read without the VSS service, deleted files, and eleven more artefacts.

Every workstation, from XP to 11

A second executable, `WAC32.exe`, built from the same sources, collects from Windows XP SP1 to Windows 8 in 32 bits; `WAC.exe` remains the program of every 64-bit version. Each version has its formats, and WAC decodes them: Prefetch and ShimCache from XP to 8.1, BITS queues, `Thumbs.db`, the WMI repository and the `.evt` logs of XP. A table of the artefacts per version decides what is taken: an artefact absent from a version is not looked for; a source that should be there and is not is declared "expected but not found" — that absence is a finding; a format without a decoder is collected raw and said "not supported", never returned as an empty list.

Shadow copies, without VSS

The hives, logs and files as each shadow copy kept them are read again from the volume itself, raw: no call to the VSS service, no write to the examined disk. What a shadow copy keeps and the volume no longer has is reported — a Run value erased since, an event of a cleared log.

Deleted files

Three levels, set in `wac.yml`: the content of the `$MFT`'s free records; the names left in the slack of the directory indexes; on demand, PE, ZIP and PDF files carved from free space at their exact size. The FAT and exFAT volumes of USB keys are read the same way. Each recovered file enters the exhibit store with its fingerprints; on an SSD, what TRIM erased is gone. On the test machine, the default level recovers 12,529 of them (4.3 GB) in 156 s.

Eleven more artefacts

The Windows search index (`Windows.edb`), Sysmon (configuration, archives) and relocated logs, PowerShell profiles, Entra ID, OneDrive, ETW files, Defender for Endpoint, the firewall log, the boot sectors (MBR, GPT, EFI partition), Intune's logs: 80 artefacts instead of 69. The manuals set this collection against MITRE ATT&CK: of the 479 Windows techniques of version 20.0, 475 leave a trace WAC collects — 463 in the logs and on the disk, 7 in the logs only, 5 on the disk only; 4 leave none.

The procedure, strengthened

The manifest's seal is printed at the end of the collection, to be written in the report, and the scripts shipped hold the manifest against it. A code and security review of the whole project came before the release.

No trace, measured again

On the frozen clone of the test machine, under Sysmon, each tool is started again from a key. WAC still writes **no file** on C: and **no registry value**; the hives, read again offline and set key by key against the frozen state, confirm it. The measure found one trace, fixed since: the module list of each process had 42 processes opened with full access by Windows; they are now opened with query and memory read only. KAPE and its modules, in the same conditions, write to the user's hive, create 8 tracing keys in SOFTWARE and 28 files on C:.

Tests on every version

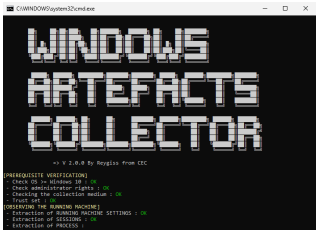
The autonomous cycle runs under Windows XP, Vista, 7, 8, 8.1 and 11. Under Windows 11, it passes 553 checks, each against an independent source; the eleven tools of Eric Zimmerman, confronted again on the same exhibits, give the same values as WAC, or wrong ones (SrumECmd shifts eight rows, AmcacheParser invents 5,500 values).

In figures

Artefacts collected (2.0: 69)	80
Versions of Windows	XP SP1 to 11
Collection without executables and deleted files (KAPE: 53 s)	64 s
Deleted files recovered, test machine	12,529
Files written on C: by WAC	0
Checks of the Windows 11 cycle	553
Windows ATT&CK techniques covered	475 / 479

Availability

Version 3.0.0, Windows XP SP1 to 11 (x86 and x64), GPL 3.0 licence: `https://github.com/dev-cec/WAC`. Manuals, validation, harmlessness and comparison files in French and English, measured on 29 and 30 October 2026.



WAC collecting.